5.3 Properties of Logarithms

Pre-Calculus 5.3 · Student sheet

Name _____ Period ____ Date _____

I can…

- use the change-of-base formula.
- expand a logarithmic expression.
- condense a logarithmic expression.

Vocabulary · fill in the missing word

Change-of-base formula — $\log_a x =$ _____ $= \dfrac{\ln x}{\ln a}$

Expand — rewrite a logarithm of a product, quotient, or power as a sum, _____, or multiple of logarithms

Condense — rewrite a sum or _____ of logarithms as a single logarithm

Part 1 · Calculator experiments

Use a calculator. Round to four decimal places.

Expression	Value	Expression	Value
$\log 2 + \log 3$		$\log 6$	
$\log 6 - \log 2$		$\log 3$	
$3 \log 2$		$\log 8$	

Part 2 · Notes and examples (fill in each note, then work the example)

Change of base: $\log_a x =$ _____ (or $\log x / \log a$)

Example 1

Use a calculator. Round to four decimal places.

a. $\log_4 25$

b. $\log_2 12$

Product and quotient: $\log_a(uv) =$ _____; $\log_a(u/v) = \log_a u - \log_a v$

Example 2

Write each in terms of $\ln 2$ and $\ln 3$.

a. $\ln 6$

b. $\ln\left(\dfrac{2}{27}\right)$

Power: $\log_a u^n =$ _____

Example 3

Find the exact value.

a. $\log_5 \sqrt[3]{5}$

b. $\ln e^6 - \ln e^2$

Example 4

Expand.

a. $\log_4 5x^3y$

b. $\ln\left(\frac{\sqrt{3x-5}}{7}\right)$

Example 5

Condense.

a. $\frac{1}{2}\log x + 3\log(x+1)$

b. $2\ln(x+2) - \ln x$

Part 3 · Partner practice

Partner 1

Expand $\ln\left(\frac{x^2y}{z}\right)$.

Partner 2

Condense $\log 4 + 2\log x - \log 3$.